
[33-120] Science and Science Fiction

Spring 2025

Instructor:	Katie Breivik	Office:	Wean 8319
E-mail:	kbreivik@andrew.cmu.edu	Office Hours:	Tuesday at 3pm or by email appointment
Class Time:	Mon/Wed/Fri 11-11:50am	Class Room:	DH A302

Course Description from Course Catalog

We will view and critique the science content in a selection of science fiction films, spanning more than 100 years of cinematic history, and from sci-fi TV shows from the past 50+ years. Guided by selected readings from current scientific literature, and aided by order-of-magnitude estimates and careful calculations, we will ponder whether the films are showing things which may fall into one of the following categories: Science fiction at the time of production, but currently possible, due to recent breakthroughs. Possible, in principle, but beyond our current technology. Impossible by any science we know. Topics to be covered include the future of the technological society, the physics of Star Trek, the nature of space and time, extraterrestrial intelligence, robotics and artificial intelligence, biotechnology and more. Success of this course will depend upon class participation. Students will be expected to contribute to discussion of assigned readings and problems.

Student Learning Outcomes

At the end of the class, students should be able to

- ☐ develop and maintain an interest in science.
- ☐ become familiar with the history of science fiction film by watching numerous short clips and full-length movies.
- ☐ develop critical thinking skills and to discern the difference between currently plausible scenarios, future possibilities, and total fantasy.
- ☐ use science fiction film as a springboard for discussing current scientific research.
- ☐ use library and/or online resources and to discern and summarize the key ideas in short scientific articles.
- ☐ work cooperatively in teams to solve challenging, but doable problems.

Textbooks and Course Materials

 **Canvas:** The site can be reached from <http://www.cmu.edu/canvas/> The course website will contain the syllabus, homework, project assignments and all required reading. Announcements and visual material presented during lectures will also be posted throughout the semester.

 **Textbooks:** Exploring Science Through Science Fiction, **Second Edition**, Author: Barry B. Loukkala, available on [Amazon](#)

Prerequisite Knowledge: No prerequisite knowledge assumed

Evaluation Procedures

The course will consist of in-class lectures and a variety of assignments, team projects, and quizzes as detailed below. There will be a final cumulative exam to be taken online on Canvas during finals week.

Course Component	Breakdown	Total Points
Science Knowledge Survey	1 on 1st day of class	25 pts
In-Class Discussion	Every class period	50 pts
Sci-fi Problem Sets	4 at 25 pts each	100 pts
Exploration Papers	5 at 25 pts each	125 pts
Quizzes on Lectures/Reading	7 w/ variable pts	350 pts
Team Projects	2 at 25 pts each	50 pts
Film Critiques	2 at 50 pts each	100 pts
Final Exam		200 pts
Total Points		1000 pts

Final grade boundaries are: A: 900 - 1000, B: 800 - 899, C: 700 - 799 pts D: 600 - 699, R: <600

Boundary cases may be discussed based on demonstrated engagement, effort, and improvement over the semester.

Attendance and Late Work Policies

Formal attendance will not be recorded for the course. However, to gain discussion points for the course, you must be present in class. Class time will also serve as a venue to communicate important details on course component, so in-class participation is likely to correlate strongly with final grade outcomes.

Late homeworks may be submitted for reduced credit at 2 pts per day up to the point that grading is complete. Assignments are allowed *one submission only* and there are **no makeup quizzes**. The dates of each quiz are listed in the course calendar and will be announced both in class and on canvas. There will be *several* opportunities for extra credit announced during class that can be used to make up points lost for a missed quiz.

Accommodations for Students with Disabilities

If you have an accommodations letter from the Disability Resources office, you are encouraged to discuss your accommodations and needs with me as early in the semester as possible. I will work with you to ensure that accommodations are provided as appropriate. If you suspect that you may benefit from accommodations but are not yet registered with the Office of Disability Resources, I encourage you to contact them at access@andrew.cmu.edu.

Statement on Diversity, Equity, and Inclusion

Everyone in this course must be treated with respect. Science is a human endeavour and we are all accountable for ensuring that the environment we create is welcoming and free from discrimination. Each of us is here for a variety of reasons, comes from a variety of backgrounds,

and have a diversity of skills and past experiences. It is expected that the students and instructor will contribute to a collaborative learning environment where diverse perspectives and approaches are celebrated.

Unfortunately, incidents may occur, whether intentional or not, that can create an unwelcoming environment for one or more class members. The university encourages anyone who experiences or observes unfair or hostile treatment on the basis of identity to speak out for justice and support, within the moment of the incident or after the incident has passed. Anyone can share these experiences using the following resources:

- Center for Student Diversity and Inclusion: csdi@andrew.cmu.edu, (412) 268-2150
- Report-It online anonymous reporting platform: reportit.net username: tartans password: plaid

All reports will be documented and deliberated to determine if there should be any following actions. Regardless of incident type, the university will use all shared experiences to transform our campus climate to be more equitable and just.

Communication Expectations

The majority of our communication will be done during class time but I am also reachable by email at the address listed at the top of this page. I will also hold a weekly office hour on Tuesdays at 2pm. I can also meet outside of those hours, but these meetings will need to be scheduled by email.

Use of Film/TV Clips Under Fair Use

We will view several clips taken from television and film in class for illustrative and critical purposes in support of the aims of the course. They will only be shown during class and not available for personal use such that all clips remain under fair use.

Student Wellness

Take care of yourself. Do your best to maintain a healthy lifestyle this semester by eating well, exercising, getting enough sleep and taking some time to relax. This will help you achieve your goals and cope with stress.

All of us benefit from support during times of struggle. You are not alone. There are many helpful resources available on campus and an important part of the college experience is learning how to ask for help. Asking for support sooner rather than later is often helpful. If you or anyone you know experiences any academic stress, difficult life events, or feelings like anxiety or depression, we strongly encourage you to seek support including from the [Timely Care App](#). Counseling and Psychological Services (CaPS) is also here to help: call 412-268-2922 and visit their [website](#).

Consider reaching out to a friend, faculty or family member you trust for help getting connected to the support that can help. If you or someone you know is feeling suicidal or in danger of self-harm, call someone immediately, day or night: CaPS: 412-268-2922 Re:solve Crisis Network: 888-796-8226

The [CMU Pantry](#) is committed to reducing hunger among students by providing nutritious food at no cost. Accessing these food resources can promote a healthier, balanced university experience.